

LINEAR REGRESSION

KJR

CONTENT

Linear Regression in Machine learning

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling



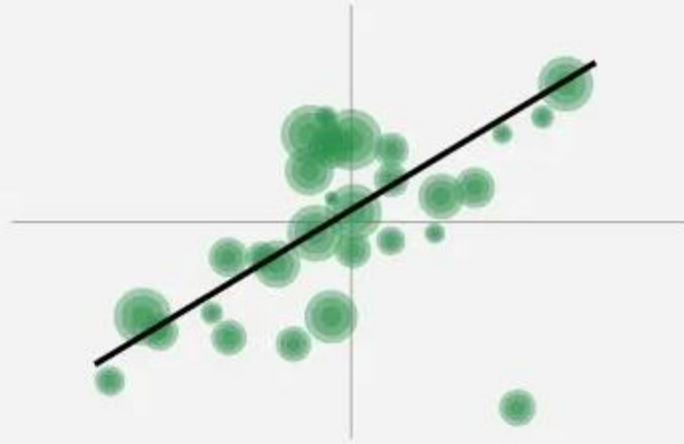
Simple model for predicting continuous values.



It finds the relationship between input features and output.



It is used in forecasting and predicting salaries based on experience.



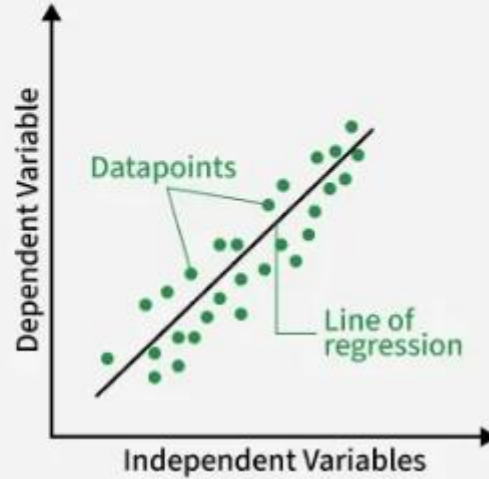
How Does Linear Regression Work?



Finds the best-fitting line by minimizing prediction errors (least squares method)



It calculates coefficients for variables that minimize the error in predictions.



For example, we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied. Here

- **Independent variable (input):** Hours studied because it's the factor we control or observe.
- **Dependent variable (output):** Exam score because it depends on how many hours were studied.

We use the independent variable to predict the dependent variable.

Best Fit Line in Linear Regression

In linear regression, the best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output). It is the line that minimizes the difference between the actual data points and the predicted values from the model.

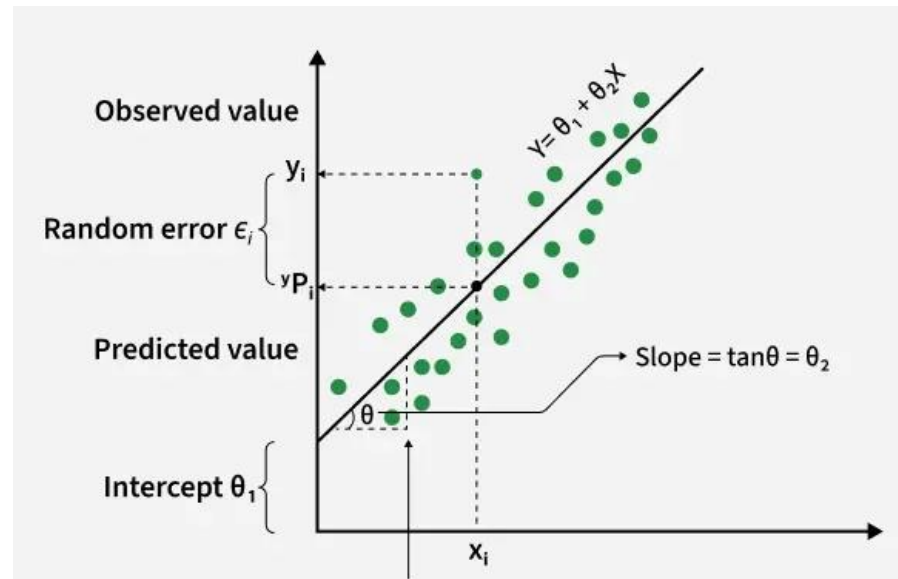
1. Goal of the Best-Fit Line

The goal of linear regression is to find a straight line that minimizes the error (the difference) between the observed data points and the predicted values. This line helps us predict the dependent variable for new, unseen data.

Here Y is called a dependent or target variable and X is called an independent variable also known as the predictor of Y .

- θ_1 represents the intercept, which is the value of Y when $X = 0$
- θ_2 represents the slope, which shows how much Y changes for a unit change in X

There are many types of functions or modules that can be used for regression. A linear function is the simplest type of function. Here, X may be a single feature or multiple features representing the problem.



2. Equation of the Best-Fit Line

For simple linear regression (with one independent variable), the best-fit line is represented by the equation

$$y = mx + b$$

Where:

- **y** is the predicted value (dependent variable)
- **x** is the input (independent variable)
- **m** is the slope of the line (how much y changes when x changes)
- **b** is the intercept (the value of y when x = 0)

The best-fit line will be the one that optimizes the values of m (slope) and b (intercept) so that the predicted y values are as close as possible to the actual data points.

3. Minimizing the Error: The Least Squares Method

To find the best-fit line, we use a method called Least Squares. The idea behind this method is to minimize the sum of squared differences between the actual values (data points) and the predicted values from the line. These differences are called residuals.

The formula for residuals is:

$$\text{Residual} = y_i - \hat{y}_i$$

Where:

- y_i is the actual observed value
- $\hat{y}_i$ is the predicted value from the line for that x_i

The least squares method minimizes the sum of the squared residuals:

$$\Sigma(y_i - \hat{y}_i)^2$$

This method ensures that the line best represents the data where the sum of the squared differences between the predicted values and actual values is as small as possible.

4. Interpretation of the Best-Fit Line

- **Slope (m):** The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x). For example if the slope is 5, it means that for every 1-unit increase in x , the value of y increases by 5 units.
- **Intercept (b):** The intercept represents the predicted value of y when $x = 0$. It's the point where the line crosses the y -axis.

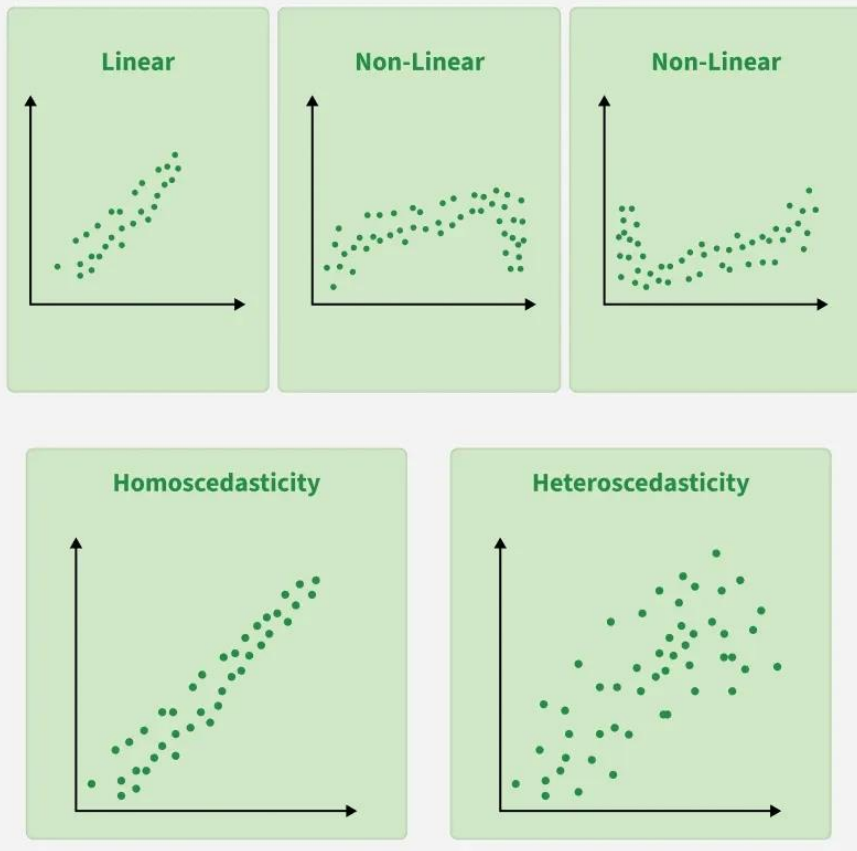
In linear regression some hypothesis are made to ensure reliability of the model's results.

Limitations:

- **Assumes Linearity:** *The method assumes the relationship between the variables is linear. If the relationship is non-linear, linear regression might not work well.*
- **Sensitivity to Outliers:** *Outliers can significantly affect the slope and intercept, skewing the best-fit line.*

Assumptions of the Linear Regression

- 1. Linearity:** The relationship between inputs (X) and the output (Y) is a straight line.
- 2. Independence of Errors:** The errors in predictions should not affect each other.
- 3. Constant Variance (Homoscedasticity):** The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.
- 4. Normality of Errors:** The errors should follow a normal (bell-shaped) distribution.
- 5. No Multicollinearity (for multiple regression):** Input variables shouldn't be too closely related to each other.
- 6. No Autocorrelation:** Errors shouldn't show repeating patterns, especially in time-based data.
- 7. Additivity:** The total effect on Y is just the sum of effects from each X, no mixing or interaction between them.'



Types of Linear Regression

When there is only one independent feature it is known as Simple Linear Regression or Univariate Linear Regression and when there are more than one feature it is known as Multiple Linear Regression or Multivariate Regression.

1. Simple Linear Regression

Simple linear regression is used when we want to predict a target value (dependent variable) using only one input feature (independent variable). It assumes a straight-line relationship between the two.

Formula

$$\hat{y} = \theta_0 + \theta_1 x$$

Where:

- $\hat{y}$ is the predicted value
- x is the input (independent variable)
- θ_0 is the intercept (value of $\hat{y}$ when $x=0$)
- θ_1 is the slope or coefficient (how much $\hat{y}$ changes with one unit of x)

Example:

Predicting a person's salary (y) based on their years of experience (x).

2. Multiple Linear Regression

Multiple linear regression involves more than one independent variable and one dependent variable. The equation for multiple linear regression is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

where:

- $\hat{y}$ is the predicted value
- $x_1, x_2, \dots, x_n$ are the independent variables
- $\theta_1, \theta_2, \dots, \theta_n$ are the coefficients (weights) corresponding to each predictor.
- θ_0 is the intercept.

The goal is to find the best-fit line that predicts Y accurately for given inputs X.

Use Cases

- **Real Estate:** Predict property prices using location, size and other factors.
- **Finance:** Forecast stock prices using interest rates and inflation data.
- **Agriculture:** Estimate crop yield from rainfall, temperature and soil quality.
- **E-commerce:** Analyze how price, promotions and seasons affect sales.

Once you understand linear regression and its types, the next step is building the model in practice.

Cost function for Linear Regression

In Linear Regression, the cost function measures how far the predicted values ($\hat{Y}$) are from the actual values (Y). It helps identify and reduce errors to find the best-fit line. The most common cost function used is Mean Squared Error (MSE), which calculates the average of squared differences between actual and predicted values:

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Here:

- $\hat{y}_i = \theta_1 + \theta_2 x_i$

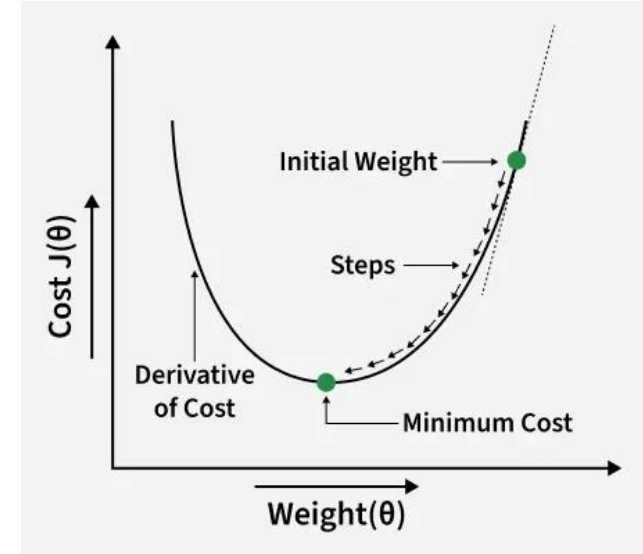
To minimize this cost, we use Gradient Descent, which iteratively updates θ_1 and θ_2 until the MSE reaches its lowest value. This ensures the line fits the data as accurately as possible.

Gradient Descent for Linear Regression

Gradient descent is an optimization technique used to train a linear regression model by minimizing the prediction error. It works by starting with random model parameters and repeatedly adjusting them to reduce the difference between predicted and actual values.

How it works:

- Start with random values for slope and intercept.
 - Calculate the error between predicted and actual values.
 - Find how much each parameter contributes to the error (gradient).
 - Update the parameters in the direction that reduces the error.
 - Repeat until the error is as small as possible.
- This helps the model find the best-fit line for the data.



Why Linear Regression is Important

Here's why linear regression is important:

- It's easy to understand and interpret, making it a starting point for learning about machine learning.
- Helps predict future outcomes based on past data, making it useful in various fields like finance, healthcare and marketing.
- Many advanced algorithms, like logistic regression or neural networks, build on the concepts of linear regression.
- It's computationally efficient and works well for problems with a linear relationship.
- It's one of the most widely used techniques in both statistics and machine learning for regression tasks.
- It provides insights into relationships between variables (e.g., how much one variable influences another).

Advantages

- Simple and easy to implement with interpretable coefficients.
- Computationally efficient, suitable for large datasets and real-time applications.
- Serves as a good baseline model.
- Relatively robust to outliers.
- Widely available in ML libraries and software.

Limitations

- Assumes a linear relationship between variables.
- Sensitive to multicollinearity among features.
- Requires proper feature engineering.
- Can overfit or underfit depending on data.
- Limited for modeling complex relationships.

Linear Regression maths

The linear regression line equation is written in the form

$$y = mx + c$$

where,

- *x is an Independent Variable, Plotted along the X-axis*
- *y is the Dependent Variable, Plotted along the Y-axis*

The intercept value, c, and the slope of the line, m, are evaluated using the formulas given below:

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$c = \frac{1}{n}(\sum y - m\sum x)$$

Solved Example Questions on Linear Regression

Question 1: Find the linear regression equation for the given data:

x	y
3	8
9	6
5	4
3	2

Solution:

Calculating intercept and slope value.

x	y	x²	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\Sigma x = 20$	$\Sigma y = 20$	$\Sigma x^2 = 124$	$\Sigma xy = 104$

Using formula,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$a = \{20 (124) - 20 (104)\} / \{4 (124) - 400\}$$

$$a = 400/96 = 4.17$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$b = \{4 (104) - 20 (20)\} / \{4 (124) - 400\}$$

$$b = 16/96 = 0.166$$

So, linear regression equation is, $y=a+bx \rightarrow y = 4.17 + 0.167x$

H.W.

1. What will be the value of y if x is 7?
2. Analysis the cost of this problem.

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Question 2: Find the linear regression equation for the given data:

x	y
4	6
7	5
3	8
1	3

Solution:
Calculating intercept and
slope value.

x	y	x ²	xy
4	6	16	24
7	5	49	35
3	8	9	24
1	3	1	3
Σx = 15	Σy = 22	Σx ² = 75	Σxy = 86

Using formula,

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$
$$= (225 - (75) 4) / ((86) 15 - (75) 22) = 360/75$$
$$= 4.8$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$
$$(225 - (75) 4) / ((22) 15 - (86) 4) =$$
$$= 14 / 75$$
$$= 0.1867$$

So, the linear regression equation is, $y = 4.8 + 0.187x$.

H.W.

1. What will be the value of y if x is 7?
2. Analysis the cost of this problem.

$$\text{Cost function}(J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

Question 3: Find the intercept of the linear regression line if $\sum x = 25$, $\sum y = 20$, $\sum x^2 = 90$, $\sum xy = 150$, and $n = 5$.

Question 4: Find the intercept of the linear regression line if $\sum x = 30$, $\sum y = 27$, $\sum x^2 = 110$, $\sum xy = 190$, and $n = 4$.

Question 5: Find the slope of the linear regression line if $\sum x = 10$, $\sum y = 16$, $\sum x^2 = 60$, $\sum xy = 120$, and $n = 4$.

Question 6: Find the slope of the linear regression line if $\sum x = 40$, $\sum y = 32$, $\sum x^2 = 130$, $\sum xy = 210$, and $n = 4$.

Question 7: Find the slope of the linear regression line if $\sum x = 50$, $\sum y = 44$, $\sum x^2 = 150$, $\sum xy = 230$, and $n = 4$.